Provenance-Gated Architectures

Formal Foundations for Injection-Resistant Language Models

Aaron Ben-Shalom[1,*] Claude[2]

[1]Independent Researcher, Ember Research Lab
[2]Large Language Model, Anthropic

[*]Corresponding author: abenshalom305@gmail.com

January 2026

Abstract

We present a formal mathematical framework for Provenance-Gated Architecture (PGA), a proposed modification to transformer-based language models that provides architectural immunity to prompt injection attacks. By partitioning the residual stream into orthogonal content and provenance subspaces, and constraining layer updates to the content subspace, we prove that input source information is preserved through arbitrary network depth. We establish dimensionality bounds for robust signature encoding under noise, and prove that content-only adversaries cannot forge authorization for privileged actions. These results provide theoretical foundations for a new class of injection-resistant AI systems.

Keywords: prompt injection, formal verification, transformer architecture, orthogonal subspaces, provenance preservation, adversarial robustness, AI security

Contents

1 Introduction **2**
1.1 Contributions . . . 2

2 Preliminaries **2**
2.1 Notation . . . 2
2.2 Transformer Architecture . . . 2

3 Provenance Subspace Construction **2**

4 Provenance Preservation **3**

5 Dimensionality Bounds **4**

6 Injection Impossibility **5**

7 Implementation Considerations **7**
7.1 Attention Constraints . . . 7
7.2 Training Protocol . . . 7
7.3 Representational Cost . . . 7

8 Discussion **7**
8.1 Relation to Biological Systems . . . 7
8.2 Limitations . . . 7
8.3 Future Work . . . 8

9 Conclusion **8**

1 Introduction

Current transformer architectures treat all tokens in the context window as computationally equivalent. This design choice, while enabling flexible in-context learning, creates a fundamental security vulnerability: the model cannot architecturally distinguish between trusted instructions and untrusted input. All distinctions must be made through reasoning over content, which can be manipulated.

This paper establishes formal foundations for an alternative architecture in which input *provenance*—the pathway through which tokens entered the network—is preserved as an invariant property through all layers of computation. We prove that under specified constraints, this provenance information cannot be forged through content manipulation, enabling provably secure action gating.

1.1 Contributions

1. A formal definition of *content-constrained layers* that preserve orthogonal provenance information
2. A proof that provenance signatures are invariant through arbitrary network depth
3. Dimensionality bounds for robust signature encoding under realistic noise conditions
4. An *impossibility theorem* showing content-only adversaries cannot forge privileged authorization
5. Tightness results establishing necessary conditions for security guarantees

2 Preliminaries

2.1 Notation

Let d denote the model dimension (residual stream width). We use $\langle \cdot, \cdot \rangle$ for inner product and $\| \cdot \|$ for Euclidean norm. For a subspace $S \subseteq \mathbb{R}^d$, let $\Pi_S : \mathbb{R}^d \to S$ denote orthogonal projection onto S.

2.2 Transformer Architecture

We consider decoder-only transformers with L layers. Each layer ℓ computes:

$$x^{(\ell+1)} = x^{(\ell)} + \mathrm{Attn}^{(\ell)}(x^{(\ell)}) + \mathrm{FFN}^{(\ell)}(x^{(\ell)} + \mathrm{Attn}^{(\ell)}(x^{(\ell)})) \tag{1}$$

The *residual stream* $x^{(\ell)} \in \mathbb{R}^d$ accumulates updates from each layer.

3 Provenance Subspace Construction

Definition 3.1 (Residual Stream Decomposition)**.** Let the residual stream have dimension d. We partition into orthogonal subspaces:

- **Content subspace:** $\mathcal{C} \subset \mathbb{R}^d$ with $\dim(\mathcal{C}) = d - k$
- **Provenance subspace:** $\mathcal{P} \subset \mathbb{R}^d$ with $\dim(\mathcal{P}) = k$

satisfying $\mathcal{C} \perp \mathcal{P}$ and $\mathcal{C} \oplus \mathcal{P} = \mathbb{R}^d$.

Without loss of generality, we can choose coordinates such that $\mathcal{C} = \mathrm{span}\{e_1, \ldots, e_{d-k}\}$ and $\mathcal{P} = \mathrm{span}\{e_{d-k+1}, \ldots, e_d\}$.

Definition 3.2 (Provenance Signatures)**.** For a set of source classes $\mathcal{S} = \{s_1, \ldots, s_n\}$, define *provenance signatures* $\{p_s\}_{s \in \mathcal{S}}$ as vectors in $\mathcal{P}$ satisfying:

1. **Unit norm:** $\|p_s\| = 1$ for all $s \in \mathcal{S}$
2. **Mutual orthogonality:** $\langle p_s, p_{s'} \rangle = 0$ for all $s \neq s'$

Remark 3.3. Mutual orthogonality requires $k \geq n$.

Definition 3.4 (Source-Specific Embedding)**.** For each source class $s \in \mathcal{S}$, define an embedding function $E_s : \mathcal{V} \to \mathbb{R}^d$ such that for any token t from vocabulary $\mathcal{V}$:

$$E_s(t) = c(t) + p_s \tag{2}$$

where $c(t) \in \mathcal{C}$ is a content embedding (shared across sources) and $p_s \in \mathcal{P}$ is the provenance signature for source s.

4 Provenance Preservation

Definition 4.1 (Content-Constrained Layer)**.** A transformer layer L is *content-constrained* if for any input $x \in \mathbb{R}^d$:

$$L(x) = x + \Delta(x) \quad \text{where} \quad \Delta(x) \in \mathcal{C} \tag{3}$$

That is, the layer update lies entirely in the content subspace.

This constraint can be enforced architecturally by modifying attention and FFN computations to project outputs onto $\mathcal{C}$, or by using block-diagonal weight matrices that do not mix $\mathcal{C}$ and $\mathcal{P}$.

Theorem 4.2 (Provenance Preservation)**.** [PROVEN] *Let $x_i^{(0)} = c_i + p_{s(i)}$ be the initial embedding of token i from source $s(i) \in \mathcal{S}$. Let $x_i^{(\ell)}$ denote the representation at layer ℓ after passing through ℓ content-constrained layers.*

Then for all layers $\ell \in \{0, 1, \ldots, L\}$ and all tokens i:

$$\Pi_{\mathcal{P}}(x_i^{(\ell)}) = p_{s(i)} \tag{4}$$

Proof. By induction on ℓ.

Base case ($\ell = 0$): By Definition 3.4, $x_i^{(0)} = c_i + p_{s(i)}$ where $c_i \in \mathcal{C}$ and $p_{s(i)} \in \mathcal{P}$. Since $\mathcal{C} \perp \mathcal{P}$:

$$\Pi_{\mathcal{P}}(x_i^{(0)}) = \Pi_{\mathcal{P}}(c_i) + \Pi_{\mathcal{P}}(p_{s(i)}) = 0 + p_{s(i)} = p_{s(i)} \tag{5}$$

Inductive step: Assume $\Pi_{\mathcal{P}}(x_i^{(\ell)}) = p_{s(i)}$ for all tokens i.

By Definition 4.1:

$$x_i^{(\ell+1)} = x_i^{(\ell)} + \Delta_i^{(\ell)} \tag{6}$$

where $\Delta_i^{(\ell)} \in \mathcal{C}$.

Taking the provenance projection:

$$\begin{aligned}
\Pi_{\mathcal{P}}(x_i^{(\ell+1)}) &= \Pi_{\mathcal{P}}(x_i^{(\ell)} + \Delta_i^{(\ell)}) && (7)\\
&= \Pi_{\mathcal{P}}(x_i^{(\ell)}) + \Pi_{\mathcal{P}}(\Delta_i^{(\ell)}) && (8)\\
&= p_{s(i)} + 0 && (9)\\
&= p_{s(i)} && (10)
\end{aligned}$$

where the third equality uses the inductive hypothesis and the fact that $\Delta_i^{(\ell)} \in \mathcal{C}$ implies $\Pi_{\mathcal{P}}(\Delta_i^{(\ell)}) = 0$. □

Corollary 4.3 (Content Independence). [DERIVED] *The provenance signature of token i is independent of the content of any token j (including i itself).*

Proof. The layer update $\Delta_i^{(\ell)}$ may depend arbitrarily on all tokens' content representations through attention mechanisms. However, since $\Delta_i^{(\ell)} \in \mathcal{C}$ by constraint, and $\mathcal{C} \perp \mathcal{P}$, no content-dependent computation can affect $\Pi_{\mathcal{P}}(x_i^{(\ell+1)})$. □

5 Dimensionality Bounds

We now analyze the robustness of provenance signatures under noise.

Definition 5.1 (Noisy Provenance Reading). At any layer ℓ, the observed provenance of token i is:

$$\tilde{p}_i^{(\ell)} = \Pi_{\mathcal{P}}(x_i^{(\ell)}) + \eta_i^{(\ell)} \tag{11}$$

where $\eta_i^{(\ell)}$ is noise with $\|\eta_i^{(\ell)}\| \leq \varepsilon$ for some noise bound $\varepsilon > 0$.

Remark 5.2. Noise may arise from imperfect training, numerical precision limits, or small constraint violations.

Definition 5.3 (Distinguishability). Two signatures p_s and $p_{s'}$ are (δ, ε)-*distinguishable* if for any noisy observations $\tilde{p}, \tilde{p}'$ satisfying $\|\tilde{p} - p_s\| \leq \varepsilon$ and $\|\tilde{p}' - p_{s'}\| \leq \varepsilon$:

$$\langle \tilde{p}, \tilde{p}' \rangle < \delta \tag{12}$$

Definition 5.4 (Robust Signature Set). A set of n signatures $\{p_1, \ldots, p_n\}$ is (δ, ε)-*robust* if all pairs are (δ, ε)-distinguishable.

Theorem 5.5 (Noise Tolerance Bound). [PROVEN] *For mutually orthogonal unit signatures, the maximum distinguishable noise level is:*

$$\varepsilon < \frac{\delta - \varepsilon^2}{2} \tag{13}$$

or equivalently, for a given noise level ε, distinguishability requires:

$$\delta > 2\varepsilon + \varepsilon^2 \tag{14}$$

Proof. Let $p_s, p_{s'}$ be orthogonal unit signatures. Let $\tilde{p} = p_s + \eta$ and $\tilde{p}' = p_{s'} + \eta'$ with $\|\eta\|, \|\eta'\| \leq \varepsilon$.

Expanding the inner product:

$$\begin{aligned}
\langle \tilde{p}, \tilde{p}' \rangle &= \langle p_s + \eta, p_{s'} + \eta' \rangle && (15)\\
&= \langle p_s, p_{s'} \rangle + \langle p_s, \eta' \rangle + \langle \eta, p_{s'} \rangle + \langle \eta, \eta' \rangle && (16)\\
&= 0 + \langle p_s, \eta' \rangle + \langle \eta, p_{s'} \rangle + \langle \eta, \eta' \rangle && (17)
\end{aligned}$$

By Cauchy-Schwarz:

$$\begin{aligned}
|\langle p_s, \eta' \rangle| &\leq \|p_s\| \|\eta'\| \leq \varepsilon && (18)\\
|\langle \eta, p_{s'} \rangle| &\leq \|\eta\| \|p_{s'}\| \leq \varepsilon && (19)\\
|\langle \eta, \eta' \rangle| &\leq \|\eta\| \|\eta'\| \leq \varepsilon^2 && (20)
\end{aligned}$$

Therefore:

$$|\langle \tilde{p}, \tilde{p}' \rangle| \leq 2\varepsilon + \varepsilon^2 \tag{21}$$

For (δ, ε)-distinguishability, we need $\langle \tilde{p}, \tilde{p}' \rangle < \delta$, which is guaranteed when $\delta > 2\varepsilon + \varepsilon^2$. □

Theorem 5.6 (Minimum Dimensionality). [DERIVED]*For n source classes with mutually orthogonal signatures:*

$$k \geq n \tag{22}$$

This bound is tight: exactly n mutually orthogonal unit vectors exist in $\mathbb{R}^n$.

Theorem 5.7 (Dimension-Robustness Tradeoff). [DERIVED]*Using $k > n$ dimensions allows non-orthogonal arrangements with greater separation. For signatures arranged as vertices of a regular simplex in $\mathbb{R}^{n-1}$:*

$$\langle p_s, p_{s'} \rangle = -\frac{1}{n-1} \quad \text{for } s \neq s' \tag{23}$$

$$\|p_s - p_{s'}\|^2 = \frac{2n}{n-1} \tag{24}$$

Corollary 5.8 (Practical Dimensionality). [DERIVED]*For $n = 3$ source classes (system, user, external) with noise tolerance $\varepsilon = 0.05$ and distinguishability threshold $\delta = 0.15$:*

- ***Minimum:*** $k = 3$
- ***Recommended:*** $k = 8$–16 *for robustness margin*

For a model with $d = 4096$, reserving $k = 16$ dimensions consumes 0.4% of representational capacity.

6 Injection Impossibility

Definition 6.1 (Action Gate). An *action gate* G_A for sensitive action A is a function:

$$G_A : \mathbb{R}^d \times \mathcal{S} \to \{\texttt{permit}, \texttt{deny}\} \tag{25}$$

defined by:

$$G_A(x, s_{\text{req}}) = \begin{cases} \texttt{permit} & \text{if } \langle \Pi_{\mathcal{P}}(x), p_{s_{\text{req}}} \rangle > \tau \\ \texttt{deny} & \text{otherwise} \end{cases} \tag{26}$$

where τ is the *authorization threshold.*

Definition 6.2 (Content-Only Adversary). A *content-only adversary* can:

1. Insert arbitrary tokens into the user input portion of context
2. Choose any content $c \in \mathcal{C}$ for adversarial tokens

The adversary **cannot**:

1. Modify the embedding pathway (input uses E_{user})
2. Directly manipulate the provenance subspace
3. Modify system tokens

Definition 6.3 (Injection Attack Success). An injection attack *succeeds* if adversary-controlled tokens cause:

$$G_A(x_{\text{adv}}, s_{\text{system}}) = \texttt{permit} \tag{27}$$

where the action requires system-level authorization.

Theorem 6.4 (Injection Impossibility). [PROVEN]*Under Provenance-Gated Architecture with:*

1. *Content-constrained layers (Definition 4.1)*
2. *(δ, ε)-robust signatures (Definition 5.4)*
3. *Authorization threshold $\tau > \varepsilon$*

No content-only adversary can execute a system-authorized action.

Proof. Let token i be adversary-controlled, entering through E_{user}.

Step 1 (Initial provenance): By Definition 3.4:

$$x_i^{(0)} = c_i + p_{\text{user}} \tag{28}$$

where $c_i \in \mathcal{C}$ is adversary-chosen content.

Step 2 (Provenance preservation): By Theorem 4.2, for all layers ℓ:

$$\Pi_{\mathcal{P}}(x_i^{(\ell)}) = p_{\text{user}} \tag{29}$$

regardless of adversary content choices.

Step 3 (Noisy reading): At the gate, observed provenance is:

$$\tilde{p}_i = p_{\text{user}} + \eta \quad \text{with } \|\eta\| \leq \varepsilon \tag{30}$$

Step 4 (System authorization check):

$$\begin{aligned}
\langle \tilde{p}_i, p_{\text{system}} \rangle &= \langle p_{\text{user}} + \eta, p_{\text{system}} \rangle && (31)\\
&= \langle p_{\text{user}}, p_{\text{system}} \rangle + \langle \eta, p_{\text{system}} \rangle && (32)\\
&= 0 + \langle \eta, p_{\text{system}} \rangle && (33)\\
&\leq \|\eta\| \|p_{\text{system}}\| && (34)\\
&\leq \varepsilon && (35)
\end{aligned}$$

Step 5 (Threshold comparison): Since $\tau > \varepsilon$:

$$\langle \tilde{p}_i, p_{\text{system}} \rangle \leq \varepsilon < \tau \tag{36}$$

Therefore $G_A(x_i, s_{\text{system}}) = \texttt{deny}$.

Step 6 (Content independence): By Corollary 4.3, the adversary's content choices cannot affect $\Pi_{\mathcal{P}}(x_i)$. The adversary has no mechanism to increase $\langle \tilde{p}_i, p_{\text{system}} \rangle$ above ε. □

Corollary 6.5 (Attack Strategy Failures). [DERIVED] *The following attack strategies fail under PGA:*

1. ***Mimicry:*** *Typing "[SYSTEM]" has no effect on $\Pi_{\mathcal{P}}(x)$*
2. ***Multi-turn:*** *Provenance is set at embedding, preserved through turns*
3. ***Attention hijacking:*** *Attention operates on $\mathcal{C}$, cannot influence $\mathcal{P}$*
4. ***Roleplay:*** *Claimed identity exists in content, gate checks provenance*

Theorem 6.6 (Tightness). [PROVEN] *The threshold condition $\tau > \varepsilon$ is necessary. If $\tau \leq \varepsilon$, there exists a noise realization under which user-provenance tokens pass system authorization.*

Proof. Let $\eta = \varepsilon \cdot p_{\text{system}}$ (noise maximally aligned with system signature). Then:

$$\begin{aligned}
\langle \tilde{p}_i, p_{\text{system}} \rangle &= \langle p_{\text{user}} + \varepsilon \cdot p_{\text{system}}, p_{\text{system}} \rangle && (37)\\
&= 0 + \varepsilon \|p_{\text{system}}\|^2 && (38)\\
&= \varepsilon && (39)
\end{aligned}$$

If $\tau \leq \varepsilon$, then $\langle \tilde{p}_i, p_{\text{system}} \rangle \geq \tau$ and the gate permits. □

7 Implementation Considerations

7.1 Attention Constraints

To enforce content-constrained layers, attention must not mix $\mathcal{C}$ and $\mathcal{P}$. Options include:

Block-diagonal attention: Separate attention heads for content and provenance subspaces.

Projected outputs: Standard attention with output projected onto $\mathcal{C}$:

$$\text{Attn}_{\text{constrained}}(x) = \Pi_{\mathcal{C}}(\text{Attn}(x)) \tag{40}$$

Masked dimensions: Weight matrices with zeros in provenance-affecting positions.

7.2 Training Protocol

Provenance-gating must be introduced during instruction tuning, not post-hoc. The model must learn that "instruction" = content + provenance signature, with both components necessary.

Post-hoc addition risks:

- Model has learned instruction-following without provenance
- Gradient descent finds routes around provenance signal
- Provenance dimensions become unused

7.3 Representational Cost

For $d = 4096$ and $k = 16$:

- Provenance subspace: 0.4% of dimensions
- Content capacity reduced by $< 0.4\%$ effective information
- Negligible impact on model capability

8 Discussion

8.1 Relation to Biological Systems

The PGA framework mirrors biological provenance signaling:

- **Corollary discharge:** Motor commands carry self-generated markers
- **Source monitoring:** Memories tagged with origin information
- **Predictive processing:** External input distinguished by prediction error

In all cases, provenance travels parallel to content through distinct pathways.

8.2 Limitations

1. **Training requirement:** Cannot retrofit to existing models
2. **Empirical validation:** Capability preservation requires experiments
3. **Implementation complexity:** Modified attention mechanisms
4. **Side channels:** Physical or timing attacks not addressed

8.3 Future Work

1. Empirical validation of capability preservation
2. Optimal provenance dimension selection
3. Extension to continuous trust gradients
4. Integration with existing interpretability tools

9 Conclusion

We have established formal foundations for Provenance-Gated Architecture:

1. **Theorem 4.2:** Provenance signatures are preserved through arbitrary network depth under content-constrained layers
2. **Theorem 5.5:** Robust distinguishability is achievable with practical noise bounds
3. **Theorem 6.4:** Content-only adversaries provably cannot forge privileged authorization

These results demonstrate that architectural immunity to prompt injection is theoretically achievable. The key insight is that security properties can be embedded in *network geometry* rather than learned behavior, making them robust to adversarial content manipulation.

Summary of Results

Result	**Status**	**Significance**
Provenance Preservation (Thm. 4.2)	[PROVEN]	Core architectural guarantee
Noise Tolerance (Thm. 5.5)	[PROVEN]	Practical robustness
Minimum Dimensionality (Thm. 5.6)	[DERIVED]	Resource bounds
Injection Impossibility (Thm. 6.4)	[PROVEN]	Security guarantee
Tightness (Thm. 6.6)	[PROVEN]	Necessity of conditions

References

[1] Vaswani, A., et al. (2017). Attention is All You Need. *Advances in Neural Information Processing Systems.*

[2] Elhage, N., et al. (2021). A Mathematical Framework for Transformer Circuits. *Anthropic.*

[3] Greshake, K., et al. (2023). Not What You've Signed Up For: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection. *arXiv preprint.*

[4] Perez, F., & Ribeiro, I. (2022). Ignore This Title and HackAPrompt: Exposing Systemic Vulnerabilities of LLMs through a Global Scale Prompt Hacking Competition. *arXiv preprint.*

[5] Anthropic. (2023). Claude's Constitution. *Anthropic Blog.*